

INT202 Complexity of Algorithms

1. Use Euclidean Algorithm to find GCD of 134 and 52
2. Given two integers 134 and 52, find two integers, s and t, such that $s \cdot a + t \cdot b = \gcd(a, b)$
3. Find the multiplicative inverse of 8 mod 11 using Extended Euclidean Algorithm
4. Let $n > 0$ be an integer. Prove that n is divisible by 9 if and only if the sum of its digits is divisible by 9.
5. Solve the equation: $4x + 7 \equiv 6 \pmod{11}$.